

Atividade Aula 12 - Ariana Ferreira dos Santos

Exercício 1

pgAdmin 4

File Object Tools Edit View Window Help

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nome-raca/postgr... x biblioteca/postgre... x carros/postgres@PostgreSQL 18* x

carros/postgres@PostgreSQL 18

Query Query History

```
21 DELIMITER ',';
22 CSV HEADER;
23
24 COPY carro (id_carro, placa, ano,
25 id_pessoa)
26 FROM 'C:/caminho/aula3_carro.csv'
27 DELIMITER ',';
28 CSV HEADER;
29
30 EXPLAIN ANALYZE
31 SELECT *
32 FROM pessoa
33 WHERE nome = 'Ana Silva';
34
35 EXPLAIN ANALYZE
36 SELECT *
37 FROM pessoa
38 WHERE nome = 'João Santos';
```

Data Output Messages Notifications

Showing rows: 1 to 6 Page No: 1 of 1

	QUERY PLAN
1	Seq Scan on pessoa (cost=0.00..1938.00 rows=6 width=23) (actual time=14.244..33.429 rows=5.00 loop...
2	Filter: ((nome)::text = 'Ana Silva')::text)
3	Rows Removed by Filter: 99995
4	Buffers: shared hit=688
5	Planning Time: 0.222 ms
6	Execution Time: 33.474 ms

Total rows: 6 Query complete 00:00:00.334

CRLF Ln 30, Col 1

POR 13:26
PTB 04/06/2026

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carros/postgres@PostgreSQL 18

Query Query History

```

21 DELIMITER ',';
22 CSV HEADER;
23
24 COPY carro (id_carro, placa, ano,
25 id_pessoa)
26 FROM 'C:/caminho/aula3_carro.csv'
27 DELIMITER ',';
28 CSV HEADER;
29
30 EXPLAIN ANALYZE
31 SELECT *
32 FROM pessoa
33 WHERE nome = 'Ana Silva';
34
35 EXPLAIN ANALYZE
36 SELECT *
37 FROM pessoa
38 WHERE nome = 'João Santos';

```

Data Output Messages Notifications

Showing rows: 1 to 6 Page No: 1 of 1

QUERY PLAN

1	Seq Scan on pessoa (cost=0.00..1938.00 rows=6 width=23) (actual time=13.292..37.630 rows=6.00 loop=1)
2	Filter: ((nome)::text = 'João Santos':text)
3	Rows Removed by Filter: 99994
4	Buffers: shared hit=688
5	Planning Time: 0.152 ms
6	Execution Time: 37.661 ms

Total rows: 6 Query complete 00:00:00.212

Successfully run. Total query runtime: 212 msec. 6 rows affected.

CRLF Ln 35, Col 1

POR 13:27
PTB 04/06/2026

Com indice

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carros/postgres@PostgreSQL 18

Query Query History

```

40 CREATE INDEX idx_pessoa_nome
41 ON pessoa (nome);
42
43 EXPLAIN ANALYZE
44 SELECT *
45 FROM pessoa
46 WHERE nome = 'Ana Silva';
47
48 EXPLAIN ANALYZE
49 SELECT *
50 FROM pessoa
51 WHERE nome = 'João Santos';

```

Data Output Messages Notifications

Showing rows: 1 to 12 Page No: 1 of 1

QUERY PLAN

1	Bitmap Heap Scan on pessoa (cost=4.34..26.73 rows=6 width=23) (actual time=0.337..0.351 rows=5.00 loops=1)
2	Recheck Cond: ((nome)::text = 'Ana Silva':text)
3	Heap Blocks: exact=5
4	Buffers: shared hit=5 read=2
5	Bitmap Index Scan on idx_pessoa_nome (cost=0.00..4.34 rows=6 width=0) (actual time=0.294..0.294 rows=5.00 loops=1)
6	Index Cond: ((nome)::text = 'Ana Silva':text)
7	Index Searches: 1
8	Buffers: shared read=2
9	Planning:
10	Buffers: shared hit=16 read=1
11	Planning Time: 3.881 ms
12	Execution Time: 0.392 ms

Total rows: 12 Query complete 00:00:00.155

CRLF Ln 50, Col 13

POR 13:30
PTB 04/06/2026

The screenshot shows the pgAdmin 4 interface with the following components:

- Object Explorer:** A tree view on the left showing the database structure. The 'carro' table is selected under the 'pessoa' schema.
- Query Editor:** The central pane shows a SQL query:


```

      37 FROM pessoa
      38 WHERE nome = 'João Santos';
      39
      40 CREATE INDEX idx_pessoa_nome
      41 ON pessoa (nome);
      42
      43 EXPLAIN ANALYZE
      44 SELECT *
      45 FROM pessoa
      46 WHERE nome = 'Ana Silva';
      47
      48 EXPLAIN ANALYZE
      49 SELECT *
      50 FROM pessoa
      51 WHERE nome = 'João Santos';
      
```
- Query History:** A tab showing the execution history of the query.
- Data Output:** A tab showing the query results. The first result is a 'QUERY PLAN' table:

Step	Operation	Cost	Rows	Width	Actual Time	Actual Rows	Actual Width	Loops
1	Bitmap Heap Scan on pessoa	(cost=4.34..26.73 rows=6 width=23)	6	23	0.306..0.325	6.00	23	1
2	Recheck Cond: ((nome)::text = 'João Santos':text)							
3	Heap Blocks: exact=6							
4	Buffers: shared hit=7 read=1							
5	-> Bitmap Index Scan on idx_pessoa_nome	(cost=0.00..4.34 rows=6 width=0)	6	0	0.264..0.265	6.00	0	1
6	Index Cond: ((nome)::text = 'João Santos':text)							
7	Index Searches: 1							
8	Buffers: shared hit=1 read=1							
9	Planning Time: 0.265 ms							
10	Execution Time: 0.380 ms							

• Qual plano foi utilizado antes do índice?

Seq Scan

• Qual plano foi utilizado após o índice?

Bitmap Heap Scan utilizando Bitmap Index Scan no índice idx_pessoa_nome.

• Houve redução no tempo de execução?

sim, caiu de ~ 16,956 ms para 0,392 ms.

• O índice foi utilizado em ambos os nomes testados? Justifique com base na saída do plano.

sim, o plano passou a utilizar o índice por de Bitmap Index Scan, que fez reduzir a quantidade de registros percorridos e melhorar o desempenho da consulta.

Exercício 2

pgAdmin 4

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carros/postgres@PostgreSQL 18

Query Query History

```

44 SELECT *
45 FROM pessoa
46 WHERE nome = 'Ana Silva';
47
48 EXPLAIN ANALYZE
49 SELECT *
50 FROM pessoa
51 WHERE nome = 'João Santos';
52
53 DROP INDEX IF EXISTS idx_pessoa_nome;
54
55 EXPLAIN ANALYZE
56 SELECT *
57 FROM pessoa
58 WHERE nascimento >= DATE '1970-01-01';

```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

QUERY PLAN
1 Seq Scan on pessoa (cost=0.00..1938.00 rows=56671 width=23) (actual time=0.028..19.898 rows=57035.00 loo...
2 Filter: (nascimento >= '1970-01-01':date)
3 Rows Removed by Filter: 42965
4 Buffers: shared hit=688
5 Planning:
6 Buffers: shared hit=14 dirtied=1
7 Planning Time: 1.522 ms
8 Execution Time: 23.723 ms

Total rows: 8 Query complete 00:00:00.144

CRLF Ln 55, Col 1

POR 13:39
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nome-raca/postgr... x biblioteca/postgre... x carros/postgres@PostgreSQL 18* x

carros/postgres@PostgreSQL 18

Query Query History

```

52
53 DROP INDEX IF EXISTS idx_pessoa_nome;
54
55 EXPLAIN ANALYZE
56 SELECT *
57 FROM pessoa
58 WHERE nascimento >= DATE '1970-01-01';
59
60 CREATE INDEX idx_pessoa_nascimento
61 ON pessoa (nascimento);
62
63 EXPLAIN ANALYZE
64 SELECT *
65 FROM pessoa
66 WHERE nascimento >= DATE '1970-01-01';

```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

QUERY PLAN
1 Seq Scan on pessoa (cost=0.00..1938.00 rows=56671 width=23) (actual time=0.031..19.363 rows=57035.00 loo...
2 Filter: (nascimento >= '1970-01-01':date)
3 Rows Removed by Filter: 42965
4 Buffers: shared hit=688
5 Planning:
6 Buffers: shared hit=19 read=1
7 Planning Time: 2.858 ms
8 Execution Time: 23.446 ms

Total rows: 8 Query complete 00:00:00.147

CRLF Ln 63, Col 1

POR 13:40
PTB 04/06/2026

• O índice criado foi utilizado pelo PostgreSQL?

Não, o PostgreSQL continuou utilizando Seq Scan.

• O plano de execução mudou após a criação do índice?

Não, permaneceu Seq Scan.

• Houve melhora significativa no tempo de execução?

Não, ficou praticamente o mesmo.

- Por que o otimizador pode optar por não utilizar um índice, mesmo quando ele existe?

A consulta retorna muitos registros, ler toda a tabela pode ser mais eficiente do que usar o índice

Exercício 3

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including tables like 'carro' and 'pessoa'. The main window shows a SQL query being executed in the 'carros/postgres@PostgreSQL 18' connection. The query is as follows:

```
60 CREATE INDEX idx_pessoa_nascimento
61 ON pessoa (nascimento);
62
63 EXPLAIN ANALYZE
64 SELECT *
65 FROM pessoa
66 WHERE nascimento >= DATE '1970-01-01';
67
68 DROP INDEX IF EXISTS idx_pessoa_nascimento;
69
70 EXPLAIN ANALYZE
71 SELECT *
72 FROM pessoa
73 WHERE nascimento >= DATE '2000-01-01'
74 AND nome = 'Ana Silva';
```

The query plan for the second query is shown below:

Step	Operation	Cost	Actual Time	Actual Rows	Actual Width	Actual Loops
1	Seq Scan on pessoa	(cost=0.00..2188.00 rows=1 width=23)	(actual time=13.428..17.987 rows=2.00 loop=1)	2	23	1
2	Filter: ((nascimento >= '2000-01-01':date) AND ((nome)::text = 'Ana Silva':text))					
3	Rows Removed by Filter: 99998					
4	Buffers: shared hit=688					
5	Planning:					
6	Buffers: shared hit=6 dirtied=1					
7	Planning Time: 1.824 ms					
8	Execution Time: 18.028 ms					

Total rows: 8 Query complete 00:00:00.162

The screenshot shows the pgAdmin 4 interface after creating the index. The SQL query is the same as in the previous screenshot, but the execution plan is different:

```
69 EXPLAIN ANALYZE
70 SELECT *
71 FROM pessoa
72 WHERE nascimento >= DATE '2000-01-01'
73 AND nome = 'Ana Silva';
74
75 CREATE INDEX idx_pessoa_nascimento_nome
76 ON pessoa (nascimento, nome);
77
78 EXPLAIN ANALYZE
79 SELECT *
80 FROM pessoa
81 WHERE nascimento >= DATE '2000-01-01'
82 AND nome = 'Ana Silva';
83
```

The query plan for the second query is shown below:

Step	Operation	Cost	Actual Time	Actual Rows	Actual Width	Actual Loops
1	Index Scan using idx_pessoa_nascimento_nome on pessoa	(cost=0.42..426.19 rows=1 width=23)	(actual time=5.299..7.840 rows=2.00 loop=1)	2	23	1
2	Index Cond: ((nascimento >= '2000-01-01':date) AND ((nome)::text = 'Ana Silva':text))					
3	Index Searches: 1					
4	Buffers: shared hit=11 read=71					
5	Planning:					
6	Buffers: shared hit=19 read=1					
7	Planning Time: 2.020 ms					
8	Execution Time: 7.870 ms					

Total rows: 8 Query complete 00:00:00.188

The screenshot shows the pgAdmin 4 interface with a query window open. The query being executed is:

```

74      AND nome = 'Ana Silva';
75
76      CREATE INDEX idx_pessoa_nascimento_nome
77      ON pessoa (nascimento, nome);
78
79      EXPLAIN ANALYZE
80      SELECT *
81      FROM pessoa
82      WHERE nascimento >= DATE '2000-01-01'
83            AND nome = 'Ana Silva';
84
85      EXPLAIN ANALYZE
86      SELECT *
87      FROM pessoa
88      WHERE nome = 'Ana Silva';

```

The query plan output shows the following details:

Step	Plan
1	Seq Scan on pessoa (cost=0.00..1938.00 rows=6 width=23) (actual time=15.430..31.161 rows=5.00 loop=1)
2	Filter: ((nome)::text = 'Ana Silva':text)
3	Rows Removed by Filter: 99995
4	Buffers: shared hit=688
5	Planning Time: 0.174 ms
6	Execution Time: 31.187 ms

A notification at the bottom right states: "Successfully run. Total query runtime: 246 msec. 6 rows affected."

- Qual plano foi utilizado antes da criação do índice composto?

Seq Scan

- Qual plano foi utilizado após a criação do índice composto?

Index Scan utilizando o índice idx_pessoa_nascimento_nome.

- O índice composto foi utilizado na consulta que filtra apenas por nome?

Não. O PostgreSQL utilizou Seq Scan.

- Por que a ordem das colunas no índice composto é relevante?

A ordem é importante porque o índice foi criado como nascimento - nome. Como a consulta usou apenas nome, o PostgreSQL não utilizou o índice.

Exercicio 4

The screenshot shows the pgAdmin 4 interface with the following components:

- Object Explorer:** Displays the database structure, including tables like `carro`, `pessoa`, and `clima_alerta`.
- Query Editor:** Contains the following SQL query:


```

      CREATE INDEX idx_pessoa_nascimento
      ON pessoa (nascimento);
      CREATE INDEX idx_pessoa_nome
      ON pessoa (nome);

      EXPLAIN ANALYZE
      SELECT *
      FROM pessoa
      WHERE nascimento >= DATE '2000-01-01'
      AND nome = 'Ana Silva';
      
```
- Query Plan:** Shows the execution plan for the query. The plan includes:
 - 1. Bitmap Heap Scan on `pessoa` (cost=4.34..26.75 rows=1 width=23) (actual time=0.112..0.119 rows=2.00 loops=1)
 - 2. Recheck Cond: `((nome)::text = 'Ana Silva')::text`
 - 3. Filter: `(nascimento >= '2000-01-01')::date`
 - 4. Rows Removed by Filter: 3
 - 5. Heap Blocks: exact=5
 - 6. Buffers: shared hit=5 read=2
 - 7. Bitmap Index Scan on `idx_pessoa_nome` (cost=0.00..4.34 rows=6 width=0) (actual time=0.079..0.080 rows=5.00 loops=1)
 - 8. Index Cond: `((nome)::text = 'Ana Silva')::text`
 - 9. Index Searches: 1
 - 10. Buffers: shared read=2
 - 11. Planning:
 - 12. Buffers: shared hit=29 read=2
 - 13. Planning Time: 3.971 ms
 - 14. Execution Time: 0.151 ms
- Data Output:** Shows the results of the query, with 14 rows returned.

• O PostgreSQL utilizou os dois índices simples na consulta?

nao, usou apenas o indice `pessoa_nome`

• Qual é o papel do operador `BitmapAnd` no plano de execução?

ele combina os resultados de dois indices para localizar apenas registros que atendem todas as condicoes da consulta

• O que o plano efetivamente fez?

usou o indice da coluna `nome` para achar os registros e depois aplicou o filtro de nascimento sobre esse registros

Exercicio 5

pgAdmin 4

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carros/postgres@PostgreSQL 18

Query Query History

```

97 EXPLAIN ANALYZE
98 SELECT *
99 FROM pessoa
100 WHERE nascimento >= DATE '2000-01-01'
101 AND nome = 'Ana Silva';
102
103 EXPLAIN ANALYZE
104 SELECT *
105 FROM carro
106 WHERE ano BETWEEN 2015 AND 2020;

```

Data Output Messages Notifications

Showing rows: 1 to 8 Page No: 1 of 1

QUERY PLAN	
text	
1	Seq Scan on carro (cost=0.00..2137.00 rows=16780 width=20) (actual time=0.032..24.343 rows=17016.00 loop=1)
2	Filter: ((ano >= 2015) AND (ano <= 2020))
3	Rows Removed by Filter: 82984
4	Buffers: shared hit=637
5	Planning:
6	Buffers: shared hit=9 dirtied=1
7	Planning Time: 0.249 ms
8	Execution Time: 25.941 ms

Total rows: 8 Query complete 00:00:00.378

CRLF Ln 103, Col 1

POR 14:00
PTB 04/06/2026

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carros/postgres@PostgreSQL 18

Query Query History

```

103 EXPLAIN ANALYZE
104 SELECT *
105 FROM carro
106 WHERE ano BETWEEN 2015 AND 2020;
107
108 CREATE INDEX idx_carro_ano
109 ON carro (ano);
110
111 EXPLAIN ANALYZE
112 SELECT *
113 FROM carro
114 WHERE ano BETWEEN 2015 AND 2020;

```

Data Output Messages Notifications

Showing rows: 1 to 12 Page No: 1 of 1

QUERY PLAN	
text	
1	Bitmap Heap Scan on carro (cost=232.29..1120.99 rows=16780 width=20) (actual time=2.211..7.152 rows=17016.00 loops=1)
2	Recheck Cond: ((ano >= 2015) AND (ano <= 2020))
3	Heap Blocks: exact=637
4	Buffers: shared hit=637 read=17
5	-> Bitmap Index Scan on idx_carro_ano (cost=0.00..228.09 rows=16780 width=0) (actual time=1.999..1.999 rows=17016.00 loops=1)
6	Index Cond: ((ano >= 2015) AND (ano <= 2020))
7	Index Searches: 1
8	Buffers: shared read=17
9	Planning:
10	Buffers: shared hit=22 read=1
11	Planning Time: 2.871 ms
12	Execution Time: 8.514 ms

Total rows: 12 Query complete 00:00:00.168

CRLF Ln 108, Col 27

POR 14:02
PTB 04/06/2026

Apos criar o indice, o Bitmap Index Scan e Bitmap Heap Scan. Isso fez reduzir o tempo de execucao: 25,9 ms para 8,5 ms.

Exercicio 6

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Query History

```

122
123 EXPLAIN ANALYZE
124 SELECT p.nome, c.placa
125 FROM pessoa p
126 JOIN carro c ON p.id_pessoa = c.id_pessoa
127 WHERE p.nome = 'Ana Silva';

```

Data Output Messages Notifications

Showing rows: 1 to 16 Page No: 1 of 1

QUERY PLAN

1	Hash Join (cost=1938.06..3837.59 rows=6 width=23) (actual time=39.941..68.743 rows=7.00 loops=1)
2	Hash Cond: (c.id_pessoa = p.id_pessoa)
3	Buffers: shared hit=1325
4	-> Seq Scan on carro c (cost=0.00..1637.00 rows=100000 width=12) (actual time=0.035..11.548 rows=100000.00 loops=1)
5	Buffers: shared hit=637
6	-> Hash (cost=1938.00..1938.00 rows=6 width=19) (actual time=34.894..34.895 rows=5.00 loops=1)
7	Buckets: 1024 Batches: 1 Memory Usage: 9kB
8	Buffers: shared hit=688
9	-> Seq Scan on pessoa p (cost=0.00..1938.00 rows=6 width=19) (actual time=16.545..34.864 rows=5.00 loops=1)
10	Filter: ((nome)::text = 'Ana Silva'::text)
11	Rows Removed by Filter: 99995
12	Buffers: shared hit=688
13	Planning:
14	Buffers: shared hit=18 dirtied=1
15	Planning Time: 2.664 ms
16	Execution Time: 68.835 ms

Total rows: 16 Query complete 00:00:00.182

CRLF Ln 127, Col 29

POR 14:22

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 - RLS Policies
 - Rules
 - Triggers
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - clima_alerta
 - clima_alerta_bd
 - dbagrins
 - nome-raca
 - postgres
 - sistema_bancario
 - Login/Group Roles
 - Tablespaces (2)
 - pg_default
 - pg_global

nome-raca/postgr... x biblioteca/postgre... x carros/postgres@PostgreSQL 18* x

carros/postgres@PostgreSQL 18

Query History

```

123 EXPLAIN ANALYZE
124 SELECT p.nome, c.placa
125 FROM pessoa p
126 JOIN carro c ON p.id_pessoa = c.id_pessoa
127 WHERE p.nome = 'Ana Silva';
128
129 CREATE INDEX idx_pessoa_nome
130 ON pessoa (nome);
131
132 CREATE INDEX idx_carro_id_pessoa
133 ON carro (id_pessoa);

```

Data Output Messages Notifications

Showing rows: 1 to 22 Page No: 1 of 1

QUERY PLAN

1	-> Bitmap Heap Scan on pessoa p (cost=4.44..4.44 rows=6 width=19) (actual time=0.000..0.002 rows=5.00 loops=1)
2	Recheck Cond: ((nome)::text = 'Ana Silva'::text)
3	Heap Blocks: exact=5
4	Buffers: shared hit=5 read=2
5	-> Bitmap Index Scan on idx_pessoa_nome (cost=0.00..4.34 rows=6 width=0) (actual time=0.075..0.075 rows=5.00 loops=1)
6	Index Cond: ((nome)::text = 'Ana Silva'::text)
7	Index Searches: 1
8	Buffers: shared read=2
9	-> Bitmap Heap Scan on carro c (cost=4.31..12.00 rows=2 width=12) (actual time=0.084..0.085 rows=1.40 loops=5)
10	Recheck Cond: (id_pessoa = id_pessoa)
11	Heap Blocks: exact=7
12	Buffers: shared hit=12 read=5
13	-> Bitmap Index Scan on idx_carro_id_pessoa (cost=0.00..4.31 rows=2 width=0) (actual time=0.027..0.027 rows=1.40 loops=5)
14	Index Cond: (id_pessoa = id_pessoa)
15	Index Searches: 5
16	Buffers: shared hit=5 read=5
17	Planning:
18	Buffers: shared hit=47 read=5
19	Planning Time: 5.816 ms
20	Execution Time: 0.253 ms
21	
22	

Total rows: 22 Query complete 00:00:00.235

CRLF Ln 123, Col 1

POR 14:23

PTB 04/06/2026

Após criar o índice , houve uma melhora absurda no tempo de execução. De 68,835 ms para 0,353 ms

Exercício 7

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (2)
 - carro
 - passoa
 - Columns
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Trigger Functions
 - Types
 - Views
 - Subscriptions
 - clima_alerta
 - clima_alerta_bd
 - dbagrirs
 - nome-raca
 - postgres
 - sistema_bancario
 - Login/Group Roles
 - Tablespaces (2)
 - pg_default
 - pg_global

nome-raca/postgr... x biblioteca/postgre... x carros/postgres@PostgreSQL 18* x

carros/postgres@PostgreSQL 18

Query Query History

```

140 AND c.ano >= 2018;
141
142 DROP INDEX IF EXISTS idx_pessoa_nome;
143 DROP INDEX IF EXISTS idx_carro_id_pessoa;
144
145 EXPLAIN ANALYZE
146 SELECT p.nome, c.placa, c.ano
147 FROM passoa p
148 JOIN carro c ON p.id_pessoa = c.id_pessoa
149 WHERE p.nascimento >= DATE '1980-01-01'
150 AND c.ano >= 2018;
151

```

Data Output Messages Notifications

Showing rows: 1 to 18 Page No: 1 of 1

QUERY PLAN
test
1 Hash Join (cost=2136.21..4371.90 rows=8486 width=27) (actual time=49.845..111.187 rows=8650.00 loops=1)
2 Hash Cond: (p.id_pessoa = c.id_pessoa)
3 Buffers: shared hit=1925
4 Seq Scan on passoa p (cost=0.00..1998.00 rows=42566 width=19) (actual time=0.037..30.631 rows=42787.00 loop=1)
5 Filter: (nascimento >= '1980-01-01'::date)
6 Rows Removed by Filter: 57219
7 Buffers: shared hit=688
8 Hash (cost=1887.00..1887.00 rows=19937 width=16) (actual time=49.534..49.536 rows=19940.00 loops=1)
9 Buckets: 32768 Batches: 1 Memory Usage: 1191kB
10 Buffers: shared hit=637
11 Seq Scan on carro c (cost=0.00..1887.00 rows=19937 width=16) (actual time=0.026..36.364 rows=19940.00 loop=1)
12 Filter: (ano >= 2018)
13 Rows Removed by Filter: 80060
14 Buffers: shared hit=637
15 Planning:
16 Buffers: shared hit=24 dircid=1
17 Planning Time: 3.125 ms
18 Execution Time: 112.448 ms

Total rows: 18 Query complete 00:00:00.173

CRLF Ln 145, Col 1

POR 14:28 04/06/2026

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (2)
 - carro
 - passoa
 - Columns
 - Constraints
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
 - Trigger Functions
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 - Subscriptions
 - clima_alerta
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 - Login/Group Roles
 - Tablespaces (2)
 - pg_default
 - pg_global

nome-raca/postgr... x biblioteca/postgre... x carros/postgres@PostgreSQL 18* x

carros/postgres@PostgreSQL 18

Query Query History

```

142 DROP INDEX IF EXISTS idx_pessoa_nome;
143 DROP INDEX IF EXISTS idx_carro_id_pessoa;
144
145 EXPLAIN ANALYZE
146 SELECT p.nome, c.placa, c.ano
147 FROM passoa p
148 JOIN carro c ON p.id_pessoa = c.id_pessoa
149 WHERE p.nascimento >= DATE '1980-01-01'
150 AND c.ano >= 2018;
151
152 CREATE INDEX idx_pessoa_nascimento
153 ON passoa (nascimento);
154
155 CREATE INDEX idx_carro_ano_id_pessoa
156 ON carro (ano, id_pessoa);
157

```

Data Output Messages Notifications

Showing rows: 1 to 26 Page No: 1 of 1

QUERY PLAN
test
1 Buffers: shared hit=600 read=0
2 Seq Scan on idx_pessoa_nascimento (cost=0.00..595.54 rows=42566 width=3) (actual time=4.481..4.482 rows=42787.00 loops=1)
3 Index Cond: (nascimento >= '1980-01-01'::date)
4 Index Searches: 1
5 Buffers: shared read=70
6 Hash (cost=1261.02..1261.02 rows=19937 width=16) (actual time=28.772..28.774 rows=19940.00 loops=1)
7 Buckets: 32768 Batches: 1 Memory Usage: 1191kB
8 Buffers: shared hit=637 read=56
9 Bitmap Heap Scan on carro c (cost=374.80..1261.02 rows=19937 width=16) (actual time=3.987..16.476 rows=19940.00 loops=1)
10 Recheck Cond: (ano >= 2018)
11 Heap Blocks: exact=637
12 Buffers: shared hit=637 read=56
13 Bitmap Index Scan on idx_carro_ano_id_pessoa (cost=0.00..369.82 rows=19937 width=0) (actual time=3.825..3.825 rows=19940.00 loop=1)
14 Index Cond: (ano >= 2018)
15 Index Searches: 1
16 Buffers: shared read=56
17 Planning:
18 Buffers: shared hit=47 read=2
19 Planning Time: 6.186 ms
20 Execution Time: 56.155 ms

Total rows: 26 Query complete 00:00:00.169

CRLF Ln 145, Col 1

POR 14:30 04/06/2026

Analise quais colunas participam do JOIN;

p.id_pessoa = c.id_pessoa

Analise quais colunas participam dos filtros da cláusula WHERE. ;

p.nascimento e c.ano

analsei as colunas no join e nos filtros da consulta, no join usei a coluna id_pessoa, que relaciona as tabelas pessoa e carro.

nos filtros foi usado as colunas nascimento e ano. Para melhorar o desempenho, criei um indice na coluna nascimneto da tabelal pessoa e um indice composto nas colunas ano e id_pessoa da tabela carro.

apos criar o indice, o postgre usou na consulta, o que fez reduzir o tempo de execucao de 112,448 ms para 56,155 ms.

Exercicio 8

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'pessoa' table. The main pane shows a SQL query window with the following SQL code:

```
149 WHERE p.nascimento >= DATE '1980-01-01'
150 AND c.ano >= 2018;
151
152 CREATE INDEX idx_pessoa_nascimento
153 ON pessoa (nascimento);
154
155 CREATE INDEX idx_carro_ano_id_pessoa
156 ON carro (ano, id_pessoa);
157
158 DROP INDEX IF EXISTS idx_pessoa_nascimento;
159 DROP INDEX IF EXISTS idx_carro_ano_id_pessoa;
160
161 EXPLAIN ANALYZE
162 SELECT *
163 FROM pessoa
164 WHERE nascimento BETWEEN DATE '1980-01-01'
165 AND DATE '1990-12-31';
166
167
```

The Query Plan window shows the execution details:

Step	Operation	Cost	Actual Time	Actual Rows	Actual Width	Actual Loops
1	Seq Scan on pessoa	(cost=0.00..2198.00 rows=15510 width=22)	(actual time=0.028..36.228 rows=15745.00 loops=1)			
2	Filter: ((nascimento >= '1980-01-01'::date) AND (nascimento <= '1990-12-31'::date))					
3	Rows Removed by Filter: 84255					
4	Buffers: shared hit=668					
5	Planning:					
6	Buffers: shared hit=9 dirtied=1					
7	Planning Time: 1.128 ms					
8	Execution Time: 38.024 ms					

Total rows: 8 Query complete 00:00:00.178

The screenshot shows the pgAdmin 4 interface. The main pane shows the same SQL query as the previous screenshot, but with the index creation and drop statements removed. The Query Plan window shows the execution details:

Step	Operation	Cost	Actual Time	Actual Rows	Actual Width	Actual Loops
1	Bitmap Heap Scan on pessoa	(cost=0.00..1235.91 rows=15510 width=22)	(actual time=4.362..8.963 rows=15745.00 loops=1)			
2	Recheck Cond: ((nascimento >= '1980-01-01'::date) AND (nascimento <= '1990-12-31'::date))					
3	Heap Blocks: exact=665					
4	Buffers: shared hit=665 read=41					
5	Bitmap Index Scan on idx_pessoa_nascimento_gist	(cost=0.00..311.38 rows=15510 width=0)	(actual time=4.194..4.195 rows=15745.00 loops=1)			
6	Index Cond: ((nascimento >= '1980-01-01'::date) AND (nascimento <= '1990-12-31'::date))					
7	Index Searches: 1					
8	Buffers: shared read=41					
9	Planning:					
10	Buffers: shared hit=60					
11	Planning Time: 2.155 ms					
12	Execution Time: 10.066 ms					

Total rows: 12 Query complete 00:00:00.164

- **Qual plano foi utilizado antes da criação do índice GiST?**

seq scan

- **O índice GiST foi utilizado após a criação?**

sim, o postgres usou o índice `idx_pessoa_nascimento_gist` por meio de `bitmap index scan`

- **Houve melhora no tempo de execução?**

sim, o tempo de execução diminuiu de 38,024 ms para 10,066 ms